Bank Management System
Database Lab Project Report

1. Introduction

Banking systems play a critical role in managing financial data securely and efficiently. With the increasing reliance on computerized systems, database-driven applications have become essential for handling customer records, transactions, and account details. This project, titled **Bank Management System**, was developed as part of the database laboratory to practically implement the concepts learned throughout the course.
The main purpose of this project is to design and implement a structured system that can store, manage, and retrieve banking information accurately while maintaining data integrity and consistency. The project focuses on real-world banking operations and demonstrates how database concepts are applied to solve practical problems.

2. Objectives of the Project

- To understand the problem domain of a banking system and identify its core entities.
- To design an efficient and normalized database using ERD and relational schemas.
- To implement SQL queries, constraints, views, and transactions.
- To apply backend validation, exception handling, and reporting mechanisms.

3. Project Overview

The Bank Management System is a database-oriented application that manages essential banking operations. It stores information related to customers, accounts, and transactions in a structured manner. The system allows users to perform operations such as creating accounts, viewing account details, depositing and withdrawing money, and generating reports.
The project was implemented using SQL for database operations and a console-based interface for user interaction. Special emphasis was placed on maintaining data accuracy, enforcing constraints, and handling invalid inputs gracefully.

Project Summary (Key Points)

- Analyzed real-world banking requirements and converted them into database entities.
- Designed ERD and database tables following normalization rules.
- Implemented SQL queries, views, stored procedures, and triggers.
- Added validation, exception handling, logging, and report generation features.

4. Evaluation Criteria Explanation (CLO-Based)

4.1 Domain Classes & Problem Understanding (CLO1)

In this project, the first step was to clearly understand the problem domain of a banking system. Core entities such as Customer, Account, Transaction, and Employee were identified. Each entity represents a real-world object and contains attributes relevant to banking operations. This step helped in building a strong foundation for the database design.

4.2 ERD & Conceptual Database Design (CLO1 & CLO2)

After identifying the domain classes, an Entity Relationship Diagram (ERD) was designed. The ERD visually represents entities, their attributes, and relationships between them. Primary keys were assigned to uniquely identify records, while relationships ensured proper linkage between customers, accounts, and transactions.

4.3 Database Tables & Constraints (CLO2)

The ERD was converted into relational tables using SQL. Appropriate constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, and UNIQUE were applied. These constraints ensured data integrity and prevented invalid or duplicate data entries in the system.

4.4 Views, Transactions & Queries (CLO2)

SQL queries were written to retrieve and manipulate data efficiently. Views were created to simplify complex queries and present data in a readable format. Transactions were used to ensure atomicity and consistency, especially during fund transfer operations.

4.5 Stored Procedures & Triggers (CLO3)

Stored procedures were implemented to automate frequently used operations such as balance updates. Triggers were used to enforce business rules automatically, for example, updating account balances after every transaction. This reduced manual errors and improved system reliability.

4.6 Backend Logic, Validation & Exception Handling (CLO3)

Backend logic was implemented to validate user inputs before processing any operation. Exception handling mechanisms were used to handle errors such as insufficient balance or invalid account numbers, ensuring that the system does not crash and provides meaningful error messages.

4.7 Logging & Error Handling (CLO4)

Basic logging mechanisms were included to record system errors and unexpected events. This helps in debugging and maintaining the system by tracking issues during runtime.

4.8 Console-Based Interaction & Output Formatting (CLO4)

The system uses a console-based interface that displays clear menus and formatted outputs. User-friendly prompts make the system easy to use, even for first-time users.

4.9 Report Generation (CLO4)

The Bank Management System supports report generation in both console and PDF formats. Reports summarize customer details, account information, and transaction history, making data analysis and record-keeping more efficient.

5. Conclusion

The Bank Management System project successfully demonstrates the practical implementation of database concepts learned in the course. By integrating ERD design, SQL queries, constraints, stored procedures, and backend logic, the project provides a complete solution to a real-world problem. The system fulfills all CLO requirements and enhances understanding of database-driven applications.

Overall, this project improved problem-solving skills, database design knowledge, and practical experience in implementing secure and reliable systems. It serves as a strong foundation for developing more advanced financial and database applications in the future.